Public Health Alliance for Genomic Epidemiology

PHA4GE Technical Report: Development of a Minimum Metadata Standard for Meningitis Surveillance (v0.1)

Executive Summary

On 24 October 2025, PHA4GE convened an in-person, community-led stakeholder workshop in Cape Town, South Africa, to co-develop a **minimum metadata standard for meningitis surveillance**, aligned with public-health use cases and the realities of low- and middle-income country (LMIC) health systems. Participants represented national reference laboratories, public health institutes, diagnostic laboratories, academic and bioinformatics expertise from South Africa, Uganda, Malawi, Burkina Faso and Senegal.

The workshop produced consensus on:

- A **tiered metadata model** consisting of compulsory data fields with recommended and optional extensions.
- **Priority public-health use cases** supported by the metadata (AMR surveillance, outbreak detection, vaccine effectiveness, and clinical guideline development).
- A decision on which **bacterial, viral, and fungal meningitis** pathogen to include within a single harmonised framework.
- Governance principles emphasising equity, feasibility, national ownership, and interoperability.

This report documents the rationale, methods, deliberations, and agreed outputs, and serves as the technical basis for piloting and further refinement of the PHA4GE Meningitis Metadata Standard (version 0.1).

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Commented [3]: this was not really sufficiently covered, but this will need to form part of a broader discussion around ethics and data sharing

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1. Background and Rationale

Genomic, molecular, and diagnostic data are increasingly central to meningitis surveillance, outbreak response, and policy decision-making. However, without accompanying contextual data (metadata), these data cannot be reliably interpreted, integrated, or compared across laboratories, public health institutions and countries or regions.

Genomic, diagnostic and molecular data require consistent metadata to maximise their utility, particularly in public health surveillance and outbreak response. Metadata fields must capture essential epidemiological, clinical, and in some cases, environmental information to enable effective monitoring and intervention. Although academics and other institutions often generate genomic data, they may not be fully aware of the critical metadata needed for public health or which fields are practically collectable.

Another critical factor is capturing metadata in a standardised way, aligned with existing ontologies. Importantly, the metadata system must accommodate diverse naming conventions to be searchable and interoperable globally.

Finally, the specification must ensure compatibility with downstream analyses and support responsible approaches to data sharing. The need for a **public-health-driven, feasible, and interoperable metadata standard** was identified as critical, particularly for LMIC contexts where infrastructure, digital maturity, and workforce capacity vary widely.

Without early engagement from public health professionals and users in low-resource settings, along with field testing, the metadata specification may end up serving only research agendas or high-income countries, thereby limiting its global applicability.

PHA4GE therefore brought together representatives with meningitis expertise across diverse settings, but with grounded public health insight to initiate the development of a minimal meningitis metadata standard.

At the inaugural meningitis stakeholder meeting held in Cape Town, **stakeholders highlighted persistent challenges including:**

- Missing or inconsistently captured clinical and epidemiological data (e.g. vaccination status, outcomes, travel history).
- Redundant and overlapping data-collection forms.
- Poor linkage between clinical, laboratory, and genomic datasets.
- Incompatibility between national reporting systems and regional or global platforms.

Commented [4]: careful in above index case and patient are different things? but are they? - we like to always call them individuals or patients

also sample and specimen are in the index - specimen is from whence the isolate/sequence comes, and in analysis you may propose to select a "sample" of the data? so not the same thing?

Commented [5]: diagnostic comes first, then molecular (may also be diagnostic) and then genomic?

Commented [6]: I think this lack of acknowledgement of importance of relevant metadata is across public health and academic institutes - it is not novel to one

Commented [7]: I always argue for clinical and then public health driven utility of genomic data - already individual level data may be needed for individual level treatment (and it will get more as sequencing gets closer to the patient and the wards), and then there is utility within surveillance and population level actions. Consider when 16S sequencing or metagenomics may help with the clinical diagnosis of a chronic meningitis case...

2. Objectives

The workshop aimed to:

1. Define the **scope and structure** of a meningitis metadata standard suitable for public health surveillance.
2. Identify **priority use cases** and **pathogens** the metadata must support.
3. Agree on **minimum, recommended, and optional metadata fields** across case, specimen, laboratory, and bioinformatics domains.
4. Agree on the type of fields (e.g. drop down, date format, free text)
5. Ensure compatibility with **WHO/AFRO, Africa CDC, and national reporting systems**.
6. Establish principles for **governance, equity, and data sharing**.

3. Methods

3.1 Workshop Design

The meeting was held as a full-day, facilitated, in-person workshop with structured sessions combining expert presentations, guided discussions, and group exercises. The meeting leveraged local and international expertise on meningitis who either routinely, or during outbreaks collect, analyse and share genomic, molecular, and/or diagnostic data and their accompanying metadata, to identify key fields for public health surveillance of meningitis. Through collaborative discussion and decision-making, participants outlined the scope, structure, and content of the minimal metadata required to build a meningitis standard suitable for public health settings.

Commented [8]: if you agree to change the order of this throughout the document - diagnosis is the first thing needed for the patient, it may be based on molecular tests but often still may be based on microscopy and culture, and genomics comes after that I think?

A skeleton of a meningitis metadata standard was prepared by PHA4GE ahead of the meeting based on data fields commonly found across WHO-, weekly TLOH-, and case-based forms. The main modules within the standard were:

i) case or patient demographics

Commented [9]: just call them patients - it is more respectful I think?

ii) specimen

iii) laboratory/diagnostic, and

iv) bioinformatics.

Stakeholders voted on whether each metadata field should be classified as required, recommended, or optional. For the designation of minimal required metadata within each

module, a consensus threshold of ≥70% agreement was applied. Fields reaching this threshold were included as required elements for that module.

3.2 Participants

Participants included representatives from South Africa, Senegal, Burkina Faso, Uganda, and Malawi, spanning public health surveillance, diagnostics, laboratory management, and bioinformatics. Both in-person and virtual participation were supported.

3.3 Data Sources

Inputs for the standard included:

- Existing national and WHO meningitis surveillance forms.
- Participant experiences with routine and outbreak data collection.
- Live annotation of a shared metadata spreadsheet, with fields classified as **Compulsory (Required), Recommended, or Optional**.

4. Scope of the Metadata Standard

4.1 Pathogens Included

Consensus was reached to include:

- **Bacterial meningitis** (*Neisseria meningitidis*, *Streptococcus pneumoniae*, *Haemophilus influenzae, Streptococcus agalactiae (GBS), Listeria monocytogenes and Escherichia coli K1, S. aureus, Klebsiella pneumoniae*).
- **Viral meningitis** (e.g. non-polio enteroviruses, arboviruses like West Nile or Usutu virus).
- **Fungal meningitis** (notably *Cryptococcus neoformans,* but also *Candida neonates*).

Commented [10]: just check this species - it is new to me: consider Candida spp. or check what species you meant. It could have been a discussion on neonatal candida meningitis?

This inclusive scope supports rational antibiotic use, improved outbreak detection, and comorbidities such as HIV-co-infections.

4.2 Tiered Model

The standard adopts a **minimum core + optional annexes** approach:

- **Compulsory fields**: feasible across all settings and required for public-health action.
- **Recommended fields**: strongly encouraged where capacity allows.

- **Optional fields**: advanced or research-oriented, included via annexes.

5. Priority Public-Health Use Cases

Participants ranked and agreed on five primary use cases:

1. **Antimicrobial resistance (AMR) surveillance**
2. **Outbreak detection and management**
3. **Vaccine effectiveness monitoring**
4. **Clinical guideline development and case management**
5. **Travel history/exposure history** (this must still be further debated)

Secondary use cases included species prevalence monitoring, serogroup/serotype tracking, and laboratory quality indicators.

These use cases directly informed the selection and prioritisation of metadata fields.

6. Agreed Metadata Domains and Key Fields

6.1 Case and Patient Context

Stakeholders prioritised a minimal set of case-level descriptors sufficient to support surveillance analyses while minimising collection burden. Fields enabling anonymised linkage, basic demographics, and temporal context were emphasised, while more detailed clinical data such as patient symptoms were generally categorised as Recommended or Optional.

Essential **eight** case-level fields included:

- Case identifier (ID)
- Country
- Reporting district/facility
- Age group
- Sex
- Date of hospital admission or reporting
- Clinical syndrome (examples can be found in dropdown)
- Vaccination status/history

Commented [11]: interesting that this came out first - for me the first use of genotyping of meningitis pathogens is to investigate if they are linked/part of an outbreak - so the second.
AMR is often tested phenotypically (and is urgently needed for treatment, so has a clinical individual level priority) and yes may be important, but is not the first priority for sequencing. Or were you prioritising nosocomial outbreak of Klebsiella?

Commented [12]: I am sorry - I need to step away, I will try to come back to finish off

Commented [13]: this may need to be further investigated because I do see this on several forms

For clinical syndrome stakeholders emphasised the need to clearly distinguish **clinical syndrome** from **symptoms** in field definitions.

6.2 Specimen and Sample Metadata

Shared identifiers were identified as critical for linking patient, laboratory, and genomic data while preserving de-identification.

Five compulsory fields were agreed for sample traceability:

- Case ID
- Sample ID
- Specimen type (e.g. CSF, blood)
- Date of sample collection
- Date received at laboratory
- Pre-antibiotic status

6.3 Diagnostic and Laboratory Metadata

The **ten** minimum diagnostic metadata included:

- Organism identified
- Gram stain result
- Culture result
- PCR result
- PCR assay/protocol
- Antigen test
- Serogroup / Serotype
- AMR results
- AST method
- AST Standard

Stakeholders strongly emphasized that linking pathogen species identification to specific antibiotics and corresponding minimum inhibitory concentration (MIC) values is essential.

6.4 Bioinformatics Metadata

Five bioinformatics fields were classified as compulsory:

- Internal Sample ID
- Sequencing technology (e.g., Illumina, ONT, PacBio, BGI, etc.)
- Assembly Software + Version
- AMR Genes Detected + Database (e.g., ResFinder)
- Key Mutations (antibiotic target genes)

Quality metrics (e.g. coverage, depth) and reference genome used were marked as recommended.

Commented [14]: I feel that this should be required but perhaps this can be debated with more meningitis experts

7. Data Integration, Reporting, and Governance

7.1 Trigger for Form Completion: Suspected vs. Laboratory-Confirmed Cases

A key point of discussion concerned whether metadata forms should be initiated at the point of clinical suspicion or only after laboratory confirmation.

In some countries (e.g., South Africa and Uganda), current systems typically trigger metadata collection only after laboratory confirmation. While this approach ensures higher data accuracy, it excludes clinically suspected cases from early surveillance datasets.

Some participants argued that capturing suspected cases could strengthen early outbreak detection by enabling symptom-based surveillance. However, concerns were raised regarding incomplete or low-quality data when laboratory confirmation is absent.

Consensus:

Suspected-case capture should be recommended, particularly to strengthen early warning systems, while laboratory-confirmed cases form the core surveillance dataset.

7.2 Reporting Levels: Responsibility Across Tiers

Ambiguity was noted regarding the responsibilities for completing metadata forms:

Discussion also highlighted variation in data entry pathways and uncertainty regarding responsibility for metadata completion. In some settings, such as Malawi, data capture

remains paper-based, whereas others, including Uganda, utilise electronic systems such as AGARI. Across contexts, clinicians often initiate forms at presentation, laboratories append organism identification and antimicrobial susceptibility data, and surveillance officers conduct follow-up and outcome verification. However, gaps in communication and feedback between these tiers were frequently reported.

National systems should clarify:

Who initiates the form

Who completes laboratory fields

Who verifies and finalizes the record

How feedback loops operate across levels

7.3 Implementation challenges and Governance Principles

Implementation challenges were closely tied to differences in digital infrastructure. For example, Uganda reported that approximately 30–40% of hospitals are digitised, illustrating the uneven transition from paper-based to electronic systems across the region. Participants also noted persistent data gaps, including missing vaccination histories and unlinked patient outcomes, which limit the analytical utility of surveillance data. Strengthening linkage between laboratory, clinical, and surveillance datasets was identified as a priority.

Alignment with Africa CDC policies and digital reporting platforms, including AGARI, was considered essential to ensure sustainability and avoid parallel reporting systems. Clear governance structures defining data ownership, custodianship, and reporting hierarchies were also viewed as foundational for long-term implementation.

Ensuring equitable participation of low-resource laboratories emerged as a cross-cutting concern. In order for metadata to be realistically implemented in public health, there must be flexibility in adoption timelines, technical assistance for metadata mapping/use, and targeted training as necessary support.

This proposed minimum metadata framework was viewed as a practical mechanism for incremental adoption regardless of digital maturity, thereby promoting inclusion while maintaining standardised reporting principles.

7.4 Examples of national reporting systems

Participants also shared examples of existing national reporting systems. South Africa operates a dual system consisting of laboratory-based email alerts alongside automated

reporting from Laboratory Information Management Systems (LIMS) to provincial surveillance officers. Senegal utilizes the Terra platform, an interoperable electronic system for notifiable diseases. Representatives from several West African countries described real-time dashboards with multi-level visibility, where outbreaks are declared by Ministries of Health upon exceeding established thresholds, with reporting timelines of approximately 24 hours. These examples illustrate both the diversity of existing systems and the importance of designing metadata standards that can integrate with established national reporting infrastructures.

8. Next Steps

PHA4GE will:

- Assign a lead to develop the standard with appropriate mapping and ontologies.
- Do another round of stakeholder engagement to verify usability and completeness of standard.
- Develop a **data dictionary** with clear field definitions.
- Pilot the standard in selected NPHIs/NPHLs.
- Refine the specification based on pilot feedback.
- Create SOP and necessary reference documents.
- Develop training materials for meningitis standard.
- Develop a **meningitis bioinformatics guidance document**

9. Conclusion

The PHA4GE Meningitis Metadata Standard represents a community-endorsed, public-health-focused approach to harmonising meningitis data across diverse settings. By prioritising feasibility, interoperability, and equity, the standard lays the groundwork for improved surveillance, faster outbreak response, and more effective policy decisions across the African region and beyond.